

A decentralised federated learning scheme for heterogeneous devices in cognitive IoT

Huanhuan Ge^a, Xingtao Yang^a, Jinlong Wang^{a,b,*}, Zhihan Lyu^c

^a School of Information and Control Engineering, Qingdao University of Technology, 266520, Qingdao, China

^b Anhui Province Key Laboratory of Intelligent Building & Building Energy Saving, Anhui Jianzhu University, Hefei Anhui, China

^c Uppsala University, Sweden

ARTICLE INFO

Keywords:

Cognitive IoT
Distributed machine learning
Federated learning
Heterogeneous devices
Knowledge transfer

ABSTRACT

Cognitive Internet of Things (IoT) technologies typically rely on substantial data collected from edge devices for data analysis and decision-making. However this reliance often leads to the inadvertent exposure of private data from smart edge devices. Federated learning (FL) is a distributed machine learning framework that protects user privacy by performing collaborative training without uploading private data. Nevertheless, applying classical FL to cognitive IoT systems to preserve privacy preservation faces significant challenges, such as central server failure and communication burden. Furthermore, when edge devices with heterogeneous data and systems participate in federated training, the learning process becomes slow and the performance of edge devices is compromised. To address these challenges, we propose a decentralised FL framework for cognitive IoT, termed DFL-MKF. In DFL-MKF, the centralised server is eliminated and each edge device is dynamically connected. We initialised models for edge devices based on computational and storage capabilities to accommodate system heterogeneity. Edge devices learned from the knowledge of multiple neighbours via knowledge transfer, and knowledge fusion was employed to aggregate the knowledge of multiple neighbours, thereby improving the performance of local models, and addressing data heterogeneity. Comprehensive experiments were performed on three image classification tasks. The results of these experiments demonstrate that the proposed method achieved superior performance compared to various baselines and improved communication efficiency.

1. Introduction

The widespread adoption of smart edge devices in the Internet of Things (IoT), such as surveillance cameras and smart glasses, has generated a substantial amount of training data for machine learning applications. Cognitive IoT (Chen, Herrera, & Hwang, 2018; Li et al., 2019; Wang, 2017), facilitated by distributed machine learning, can integrate data from these edge devices for analysis and mining, enabling more intelligent decision making and enhanced services. However, during this process, sensitive data from edge devices may be leaked (Shi, Zhang, Fan, Ma, & Guan, 2023; Vellingiri et al., 2023), posing significant privacy and security risks to the public in practical applications. Federated learning (FL) (McMahan, Moore, Ramage, Hampson, & y Arcas, 2017) offers a solution by enabling edge devices to collaboratively train a shared global model under the coordination of a centralised server without transmitting private data. This approach has garnered considerable attention in the context of cognitive IoT. As shown in Fig. 1(a), when heterogeneous edge devices perform

image classification tasks, each edge device performs local training and uploads model parameters to the central server. The central server aggregates these model parameters and broadcasts the updated global model to all edge devices during each communication round. Consequently, classical FL provides a privacy-preserving mechanism (Islam, Ghasemi, & Mohammed, 2022; Kairouz et al., 2019; Liu et al., 2024; Rasha, Li, Huang, Gu, & Li, 2023; Tan, Yu, Cui, & Yang, 2022) for edge devices in the context of cognitive IoT.

PFedEff (Shi et al., 2023) proposes a personalised FL framework with efficient communication in cognitive IoT, aiming to reduce communication volume and maintain training accuracy. Khan et al. (2020) proposes an FL framework for resource optimisation of cognitive devices. However, several challenges hinder the application of FL methods in cognitive IoT environments, primarily owing to centralised servers and system and data heterogeneity (Beltrán et al., 2022; Li, Sahu, Talwalkar, & Smith, 2020). Traditional FL relies on a central server, which

* Corresponding author at: School of Information and Control Engineering, Qingdao University of Technology, 266520, Qingdao, China.

E-mail addresses: ggehuanhuan@163.com (H. Ge), yangxingtao@stu.qut.edu.cn (X. Yang), qdwangjinlong@163.com (J. Wang), lvzhihan@gmail.com (Z. Lyu).

<https://doi.org/10.1016/j.ijcce.2024.08.001>

Received 21 January 2024; Received in revised form 28 July 2024; Accepted 6 August 2024

Available online 8 August 2024

2666-3074/© 2024 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

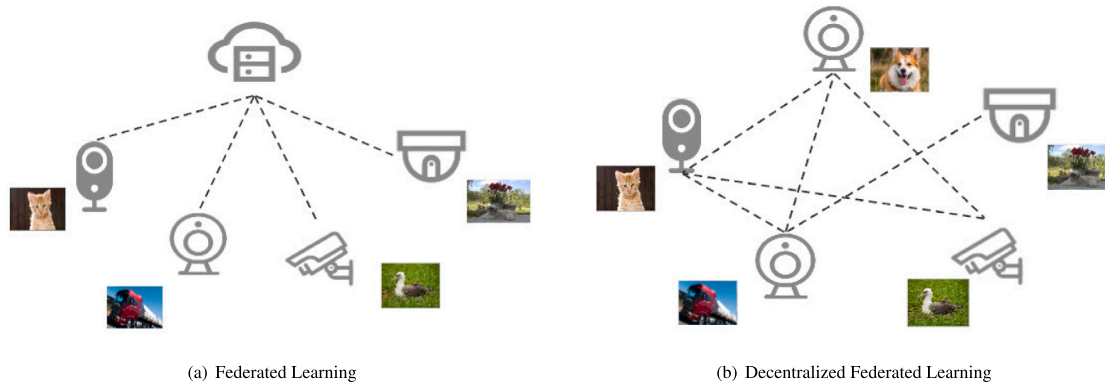


Fig. 1. Federated Learning vs Decentralised Federated Learning.

is vulnerable to attacks, creates a communication burden and introduces a single point of failure. This centralised architecture necessitates that each edge device communicates with the central server, which limits its applicability in certain scenarios. In addition, edge devices in cognitive IoT can originate from various users or institutions, which leads to highly unequal or non-independent and non-independent and identically distributed (non-IID) data distributions. Furthermore, system heterogeneity indicates that edge devices in cognitive IoT may vary in terms of their computational capabilities and storage capacities, resulting in imbalanced training times.

Recent studies (Dai, Shen, He, Tian, & Tao, 2022; Li, Zhou, Tian & Tao, 2022) have employed decentralised architecture to mitigate the adverse effects of centralised servers. Compared with centralised FL, decentralised FL (Fig. 1(b)) is a promising framework that eliminates the need for a central server. Edge devices connect dynamically and communicate with each other in a peer-to-peer manner. Despite an increasing number of participants, the learning process maintains optimal performance. Consequently, decentralised FL reduces the communication burden among edge devices, mitigates the risk of central server failures and enhances robustness. In addition, previous studies (He, Annaram, & Avestimehr, 2020; Liu, Li, Shao, & Wang, 2022; Zhu, Hong, & Zhou, 2021) have demonstrated that knowledge transfer can significantly alleviate the impact of heterogeneous data in FL. Several studies (Bhuyan & Moharir, 2022; Li, Li, Zhao, Zhang, & Li, 2021; Muhammad, Lin, Gao, & Chen, 2022) have demonstrated that FL with multiple models can effectively adapt to system heterogeneity and reduce the training costs of edge devices. However, these methods typically focus on optimisation strategies for either the centralised server or data and system heterogeneity; they lack a unified framework that comprehensively addresses all three aspects.

This study addresses the abovementioned challenges by proposing a novel decentralised FL framework for heterogeneous devices, termed DFL-MKF. The proposed framework leverages knowledge transfer and fusion to aggregate knowledge from edge devices using different models. We replaced the centralised server with a decentralised network topology. To accommodate device heterogeneity, we designed multiple models for edge devices in cognitive IoT rather than using a unified global model, which varies in structure or the number of parameters. In addition, we used knowledge transfer and fusion to aggregate knowledge from multiple neighbouring devices, which mitigated the impact of heterogeneous data in FL and cognitive IoT settings. Compared to existing methods, the proposed method offers superior performance while providing a mechanism to protect privacy. This mechanism addresses the issue of privacy leakage in real-world scenarios, thereby ensuring the security of public data.

Below, we summarise the primary research motivations for this study:

(1) Most existing FL methods in cognitive IoT rely on a central server, which introduces the risk of privacy leakage. Therefore, it is

important to develop an FL framework that operates independently of centralised servers to mitigate such privacy concerns.

(2) The heterogeneity of storage and computing capabilities of edge devices in cognitive IoT pose a significant challenge that has yet to be effectively addressed.

(3) The performance of FL in cognitive IoT is significantly affected by data heterogeneity among edge devices. However, existing methods to improve performance have demonstrated limited effectiveness in addressing this issue.

Considering these research motivations, the main contributions of this study are as follows:

(1) To address the issue of privacy data leakage, we propose a novel decentralised FL framework called DFL-MKF. The proposed framework provides an effective mechanism to protect the privacy of edge devices in cognitive IoT.

(2) The proposed method offers an efficient multimodel training approach specifically designed for heterogeneous edge devices. It effectively addresses system heterogeneity and enhances the communication efficiency of FL.

(3) Using knowledge transfer for model aggregation, effectively mitigates the impact of data heterogeneity. Extensive experiments demonstrated that the proposed algorithm outperforms existing baselines, achieving superior performance.

The remainder of this manuscript is organised as follows: Section 2 elaborates on the background knowledge and literature related to cognitive IoT and FL. In Section 3, we describe the problem under consideration and explain the DFL-MKF algorithm in detail. This section explains the approach to use the proposed DFL-MKF algorithm for multimodel federated training via knowledge transfer and fusion, thereby enhancing the performance of edge devices. Section 4 presents extensive experiments, including performance metrics, a comparison of communication efficiency and the impacts of heterogeneity, neighbours, hyperparameters α and local epochs E . These experiments aimed to demonstrate the superiority of the proposed DFL-MKF algorithm to baseline methods. Finally, Section 5 concludes the manuscript and discusses directions for future research. Table 1 provides a full-form reference for the abbreviations used in this manuscript.

2. Related work

2.1. Cognitive IoT

Cognitive IoT (Wu et al., 2014) is an advanced IoT system that realises intelligent reasoning and decision making (Chen et al., 2018) by simulating human cognitive processes by integrating technologies, such as big data analysis and distributed machine learning. However, cognitive IoT systems require extensive data for model training. During this process, there is a potential risk of privacy data leakage from edge devices, raising public concern about privacy security.

Table 1
List of abbreviations.

Abbreviation	Full Name
FL	Federated Learning
IoT	Internet of Things
Non-IID	Non-Independently Identically Distributed
CMPM	Cognitive-based Middleware for Private data Mashup
EEG	Electroencephalogram
FedAvg	Federated Averaging
QSFL	Qualification judgment & Segment for FL
L2C	Learning to Collaborate
DisPFL	PFL via Decentralised Sparse Training
Def-KT	Decentralised FL via mutual Knowledge Transfer
FedAD	Attention Distillation for FL
FedGen	Federated Distillation via Generative Learning
RMMFL	Robust Multi-Model FL
KT	Knowledge Transfer

Various methods have been proposed to address this challenge. For example, CMPM (Elmisery, Sertovic, & Gupta, 2017) proposes a middle-ground approach with hiding mechanisms to safeguard privacy in merged datasets in mashup applications. A method (Schiliro, Moustafa, & Beheshti, 2020) using normalised correlation analysis was proposed to protect EEG data models from data leakage. According to Grissa, Yavuz, and Hamdaoui (2017), a scheme is proposed to protect the location privacy of secondary users for database-driven cognitive radio networks. However, most existing approaches fail to provide effective solutions to address the data and system heterogeneity that exists in edge devices in cognitive IoT environments.

2.2. Federated learning

Traditional FL (McMahan et al., 2017) is a collaborative learning framework designed to address the problem of data islands and protect the privacy of data on edge devices. In this framework, only model training parameters, not private data (Li et al., 2020; McMahan et al., 2017), are transmitted between edge devices and a central global server. Therefore, the FL mechanism is particularly well-suited to cognitive IoT applications and can effectively prevent the leakage of sensitive data.

However, classical FedAvg (McMahan et al., 2017) suffers from client drift, which leads to poor performance in local models when the data of edge devices in cognitive IoT are heterogeneous (Karimireddy et al., 2020; Lu, Pan, Dai, Si, & Zhang, 2024). To address the impact of non-IID data in classic FL, many variants of FedAvg (Karimireddy et al., 2020; Li, Hu, Beirami & Smith, 2021; Zhang, Wang, Zhou, Wu, & Zhan, 2021) have been proposed. Ditto Li, Hu, et al. (2021) mitigates the impact of non-IID data on the global model by adding constraints. QSFL (Yi, Gang, & Xiaoguang, 2022) and the study by Yang, Zhou, Wen, Zhou, and Zhang (2023) selected high-quality edge devices to participate in training to alleviate client drift. Some approaches (Ma, Zhang, Guo, & Xu, 2022; Zhang, Guo, et al., 2021) address data heterogeneity using personalised local models for each edge device with heterogeneous data. However, these methods are limited to centralised FL settings and cannot overcome the issue of centralised server failures. Decentralised FL (Beltrán et al., 2023; Gabrielli, Pica, & Tolomei, 2023; Gao, Thai, & Wu, 2023; Han et al., 2024; Witt, Heyer, Toyoda, Samek, & Li, 2023; Zong, Qin, Wu, Tang, & Li, 2024) offers significant advantages, such as fault tolerance, and addresses the issue of centralised server failures. Recent studies have proposed methods to improve decentralised FL. For example, L2C (Li, Zhou et al., 2022) enhances the performance of local models on each device by dynamically updating the blending weights to minimise the local loss in each edge device. DisPFL (Dai et al., 2022) employs a binary mask to perform sparse training, thereby accelerating communication in decentralised FL. However, these methods fail to adapt to system heterogeneity owing to the storage requirements of edge devices in cognitive IoT environments.

Table 2
All symbols in DFL-MKF.

Symbol	Descriptions
K	The number of edge devices in cognitive IoT.
D_k	The private dataset owned by the k th device.
E	The number of epochs of device's local training.
B	The batch size of device's local model.
N	The total number of samples.
S	The number of neighbours.
T	The total numbers of communication rounds.
θ	The weight of local model in decentralised FL.
η	The learning rate of the local model.
C	The number of classes of data samples.
α	The hyper-parameter of knowledge transfer.
p	The logit produced of the model.
m	The weight of computational capability of edge device.
n	The weight of storage capability of edge device.

2.3. Knowledge transfer

Knowledge transfer (KT) (Hinton, Vinyals, & Dean, 2015) accelerates the training of small models and enhances their performance by compressing and integrating the knowledge from a single large model into a smaller model. Recent methodologies have suggested that KT can be a viable solution to address the issue of data heterogeneity while adapting to the system heterogeneity of edge devices in cognitive IoT.

Def-KT (Li, Li, & Varshney, 2022) effectively addresses the issue of imbalanced data distribution in decentralised FL via knowledge transfer. However, this method uses a ring-shaped topology, which is not suitable for the general decentralised FL scenario. FedAD (Gong et al., 2021) applies attention distillation to natural and medical fields, consuming fewer communication resources and improving the prediction performance of medical images. However, this approach requires a public dataset, which poses a risk of data privacy leaks. FedGen (Zhu et al., 2021) addresses heterogeneous FL through data-free KT, where the global server learns a lightweight generator to integrate user information in a data-free manner and broadcasts it to edge devices. RMMFL (Muhammad et al., 2022) proposes a multimodel distillation method, enabling each local model to enhance its performance and generalisability by fusing knowledge from other models. Although most of these methods are used in centralised FL, KT and multimodel training have not been widely discussed in cognitive IoT. Therefore, this study proposes a decentralised FL framework that combines KT and multimodel training methods to protect user privacy in cognitive IoT environments.

3. Methodology

This section describes the proposed DFL-MKF algorithm and comprehensively elucidates the approach to decentralised FL via KT and multimodel training in cognitive IoT settings. We first provide an overview of the complete DFL-MKF framework. Then, we furnish pertinent definitions and formulas pertaining to decentralised FL, knowledge transfer and fusion in the context of cognitive IoT. Finally, we describe the DFL-MKF algorithm and elaborate on the entire process. The symbols used in this manuscript are listed in Table 2.

Fig. 2 shows the process flow of the proposed algorithm, which employs KT and fusion for multimodel training in decentralised FL. The operations executed by each edge device with its specific model in each communication round are summarised as follows: The edge device (1) randomly selects multiple other devices as neighbours and receives models from these selected neighbours, (2) obtains logits from neighbours via KT, (3) aggregates the logits of neighbours into a teacher logit through knowledge fusion, (4) updates its local model based on the teacher logit and (5) broadcasts the updated model to the required neighbours.

3.1. Preliminaries and definitions

In the context of cognitive IoT, there are K edge devices in a dynamically connected network. We denote the k th device by $k \in \{1, 2, \dots, K\}$. Each edge device has a private dataset with image labels $D_k = (x_i^k, y_i^k)_{i=1}^{N_k}$, where x_i^k and y_i^k represent the i th sample and label of the k th device's private data respectively. N_k represents the total number of samples owned by the k th device.

Optimisation objective of traditional FL are usually based on global model. However, when the data distribution is not independent and identically distributed, the global model may cause serious accuracy degradation on edge devices. Therefore, we design personalised model training for heterogeneous edge devices, and set the optimisation goal as the local model of the edge device. We define the following objective function for each edge device in decentralised FL as Eq. (1):

$$\min_{\theta^k} f(\theta^k) = \sum_{i=1}^{N_k} L(\theta^k, (x_i^k, y_i^k)). \quad (1)$$

where L is the local loss of the edge device and θ^k is the model parameters of the edge device k .

To deploy personalised models for edge devices in cognitive IoT and enhance performance in scenarios with heterogeneous data, we incorporate knowledge transfer and fusion into the settings of decentralised FL. In traditional knowledge transfer, the edge device acquires the local logit from the local model and learns from the teacher model to obtain the teacher logit. By reducing the difference between teacher logit and local logit, the local model of the edge device gains more knowledge, thereby improving the performance of edge devices. We employ KL to measure the difference in logits, as shown in Eq. (2):

$$KL(p(\theta^k), p(\theta^T)) = \sum_{c=1}^C p_c(\theta^T) \log \frac{p_c(\theta^T)}{p_c(\theta^k)}. \quad (2)$$

where KL is Kullback Leibler Divergence, which is commonly used to measure the distance between two probability distributions in machine learning. $p(\theta^k)$ and $p(\theta^T)$ are logits which are generated by edge device k and teacher respectively. p_c is the probability that the sample belongs to the c th class. Specifically, we show the calculation formula of p_c as Eq. (3):

$$p_c = \frac{\exp(z_c)}{\sum_i \exp(z_i)}. \quad (3)$$

where z_c is the output of local model corresponding to the c th class.

Based on the above introduction and definition, we employ the knowledge fusion to extend a single teacher model in knowledge transfer into multiple ones. This enables edge devices in cognitive computing to concurrently learn the knowledge of multiple neighbours, as shown in the following Eq. (4):

$$p(\theta^T) = \sum_{i \in S_k} \frac{N_i}{\sum_{k \in S_k} N_k} p^i. \quad (4)$$

where $p(\theta^T)$ is the fused teacher logit used to guide the model training of local edge device. p^i is the logit generated by the i th neighbour, S_k is all neighbours of the edge device k , and N_i is the number of the training samples owned by the i th edge device.

Therefore, we combine the local training loss of the edge device in decentralised FL and the loss of knowledge transfer as the final loss function, the objective function of the edge device k can be written as Eq. (5):

$$\min_{\theta^k} f(\theta^k) = \sum_{i=1}^{N_k} L(\theta^k, (x_i^k, y_i^k)) + \alpha KL(p(\theta^k), p(\theta^T)). \quad (5)$$

where α is a hyper-parameter used to represent the weight of the knowledge transfer loss in the optimisation objective.

Algorithm 1 DFL-MKF

1: **Input:** Total number of devices K ; Total communication rounds T ; Local training rounds E ; The private data of device D ; The number of neighbours S ; The learning rate of the local model η ; The parameter of KT α .

2: **Initialisation:** All devices initialise different model θ^k according to computational capability m and storage capability n .

3: **Output:** Local model θ_{t+1}^k .

4: **for** $t=0, \dots, T-1$ **do**

5: **for** device $k=0, \dots, K-1$ **do**

6: $S_k \leftarrow$ select neighbours randomly from K devices.

7: **for** each device $i \in S_k$ **do**

8: $p^i \leftarrow \text{model}(D_k, \theta^i)$

9: **end for**

10: $p(\theta^T) \leftarrow \sum_{i \in S_k} \frac{N_i}{\sum_{k \in S_k} N_k} p^i$

11: $\theta_{t+1}^k \leftarrow \theta_t^k - \eta \frac{\partial (L(p(\theta^k), y) + \alpha KL(p(\theta^T) || p(\theta^k)))}{\partial \theta_t^k}$

12: Broadcast θ_{t+1}^k to other devices.

13: **end for**

14: **end for**

3.2. Proposed DFL-MKF

As mentioned previously, applying traditional KT to cognitive IoT presents three major problems: centralised server failures and data and system heterogeneity. These issues can lead to system failure, performance degradation and reduced training efficiency.

To address these concerns, we considered adopting a fully decentralised topology in cognitive IoT, which eliminates the risk of server failures and improves robustness. In addition, each edge device is dynamically connected to other devices and communicates in a peer-to-peer manner. To alleviate the communication burden on edge devices, we propose a strategy in which each device randomly selects a fixed number of devices, rather than all devices, as neighbours in each communication round. After each round of federated training, edge devices can choose to keep or replace neighbours.

Furthermore, we consider the scenario of system heterogeneity. By designing multimodel training, the proposed method enables edge devices with superior computational capabilities to process more complex models. Conversely, edge devices with limited computing capabilities can still effectively participate using lightweight models, thus reducing federated training costs. To efficiently organise multimodels for federated training and mitigate the impact of data heterogeneity, we incorporated KT and fusion rather than simply averaging the model parameters of each neighbour. During KT, each edge device shares its knowledge with selected neighbours. We employ logit fusion which uses the proportion of the number of samples as weights to integrate knowledge from multiple neighbours. This approach mitigated the adverse impact of data heterogeneity and enhanced the performance of edge devices in cognitive IoT environments.

Algorithm 1 provides a comprehensive depiction of the entire workflow involved in the DFL-MKF algorithm. We began by extending centralised FL to decentralised FL in cognitive IoT. In the centralised FL, if the central server fails, the federated training process cannot proceed normally. In decentralised FL, if an edge device cannot communicate with a faulty neighbour, it could choose to communicate with other neighbours. Therefore, decentralised FL provides a robust fault-tolerance mechanism. In addition, we addressed the challenge of system heterogeneity by initialising local models for each edge device. These local models can vary in network structure (line 2). Specifically, we assigned weights m and n based on the computational and storage capacities of each device. The sum of m and n represents the overall device capability. We sorted edge devices according to this sum and allocated models accordingly. In each round of communication t , the

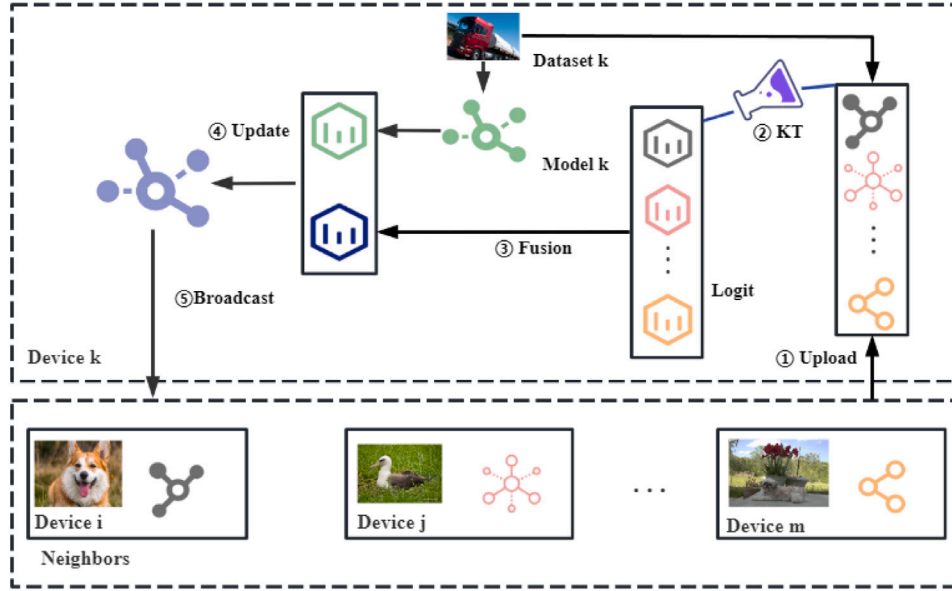


Fig. 2. The framework of DFL-MKF.



Fig. 3. Data samples on EMNIST, CIFAR10 and CIFAR100 datasets.

edge device k randomly selected devices S as neighbours to acquire knowledge. In this case, the edge device k only needed to communicate and exchange models with its neighbours, rather than all edge devices. Consequently, the communication burden on the edge device k was significantly reduced.

In order to solve subpar performance caused by data heterogeneity in decentralised FL, we incorporate the logit transfer and fusion into our proposed algorithm. Specifically, line 7 to line 9 obtains the logit p^i generated by the model of each neighbour on private local data D_k . In line 10, we perform logit fusion by combining the logits from multiple neighbours as in Eq. (4). We can observe that DFL-MKF employs a weighted average based on the size of the data N to obtain the teacher logit instead of a simple averaging, which accurately reflect the contribution of each neighbour model. In contrast, the simple average fails to differentiate the contribution differences among different neighbour models, which may result in inaccurate evaluations of the knowledge contained in neighbours. Therefore, the weighted average approach utilised by DFL-MKF can more accurately fuse and utilise the information from each neighbour model, thereby enhancing the performance in accuracy and generalisation ability of local models. Finally, we update the model of the edge device in line 11, specifically, $L(p^k, y)$ represents the training loss of the local model, which is generally obtained by calculating the loss of cross entropy. $KL(p^T \| p^k)$ represents the divergence between the teacher logit and the local logit, and α is a hyper-parameter which denotes the weight of the loss of KL .

In summary, the proposed model makes significant contributions to extending decentralised FL for multimodel knowledge fusion in cognitive IoT:

- (1) We adapted a fully decentralised topology to a cognitive IoT setting, addressing the risk of system failure caused by the centralised servers in traditional FL and improving the robustness of federated training.
- (2) We proposed an innovative design that used multimodel training in the decentralised FL process. The proposed method effectively leveraged the advantages of different edge devices and enhanced the communication efficiency of the entire federated training process.
- (3) Using KT and fusion, edge devices can acquire knowledge from other devices, which alleviates the prediction bias and generalisability degradation caused by data heterogeneity.
- (4) Unlike existing KT methods in FL settings, the proposed algorithm eliminates the need for public datasets, thereby providing enhanced user privacy protection in cognitive IoT applications.

4. Experiments

In this section, we describe comprehensive experiments conducted on three image classification datasets to assess the performance of the proposed DFL-MKF algorithm in the context of decentralised FL. This evaluation includes accuracy metrics, a comparative analysis of communication efficiency and the impacts of heterogeneity, number of neighbours, hyperparameter α and local epochs E .

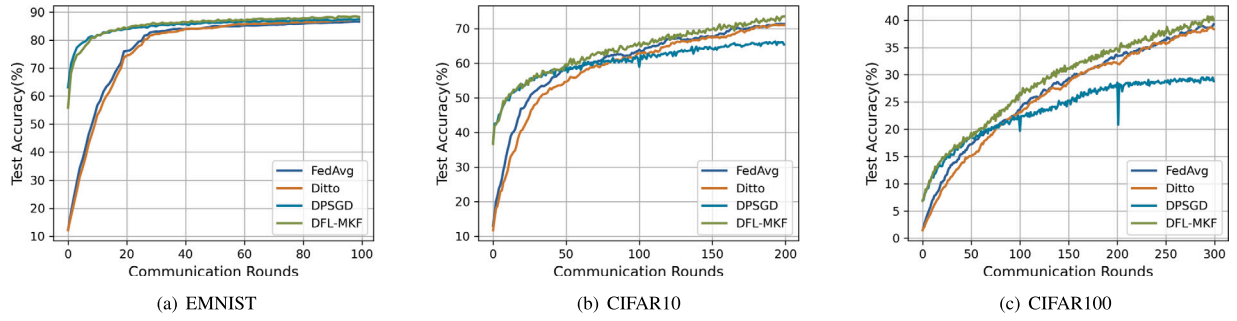


Fig. 4. Performance comparison of four algorithms on EMNIST, CIFAR10 and CIFAR100 datasets.

Table 3

The table describes the details of hardware configuration of all experiments.

Name	Value
CPU	Intel(R) Core(TM) i7-11800H @ 2.30GHz
GPU	NVIDIA GeForce RTX 3060
Memory	32GB
Graphics memory	16GB

Table 4

The table describes the settings of parameters in all experiments.

Parameter	Symbol	Value
The total number of devices	K	100
Communication rounds	T	100
The round of local training	N	10
Learning rate	η	0.01
Batch size	B	64
Hyper-parameter of KT	α	0.5

4.1. Experimental setup

4.1.1. Experimental environment

All experiments were conducted using the PyCharm integrated development environment with experimental code written in Python 3.6. The experiments were run on the Ubuntu 18.04 operating system, using the PyTorch (version 1.9.0) deep learning framework to train the model. The hardware configurations related to this experiment are listed in Table 3.

4.1.2. Datasets

For the experiments, we used three widely used image classification datasets: **EMNIST**, **CIFAR10** and **CIFAR100**. The EMNIST dataset (Cohen, Afshar, Tapson, & Van Schaik, 2017) is an extension of the MNIST dataset and includes 62 labels, consisting of 10 numbers, 26 lowercase letters and 26 uppercase letters. Each image was 28×28 pixels in size. The CIFAR10 and CIFAR100 datasets (Krizhevsky et al., 2009) include a total of 60,000 images, with 50,000 samples in the training and 10,000 samples in the test set. Each image was 32×32 pixels in size. The CIFAR10 and CIFAR100 datasets comprised 10 and 100 classes, respectively. Data samples from the EMNIST, CIFAR10 and CIFAR100 datasets are shown in Fig. 3.

To simulate non-IID data for edge devices, we employed the Dirichlet distribution to model the data distribution across these devices. We used nonuniform sampling for each class in the dataset and distributed the sampled data to each edge device participating in the federated training. The Dirichlet distribution used $\text{Dir}(\beta)$ to represent the unequal distribution of data among edge devices, where β is a hyperparameter indicating the degree of non-IID. A large value of β indicates a more uniform data distribution among edge devices and higher degree of non-IID. In contrast, a smaller value of β indicates a greater skew in the data distribution of the device and lower degree of non-IID. In the default setting, we set the hyperparameter β to 0.3.

4.1.3. Model structure

In the proposed algorithm, we customised specific models for each edge device according to its computational and storage capabilities to accommodate system heterogeneity. Specifically, we varied the computational capabilities of the edge device m and storage capacity n within the range [1, 2, 3, 4, 5] and allocated edge devices evenly to different models based on the sum of m and n . For the EMNIST dataset, we allocated either an MLP model or one of two CNN models to each edge device with heterogeneous resources. The MLP model has a hidden layer, and the first CNN model has two convolutional and two fully connected layers. The second CNN model comprises three convolutional layers and two fully connected layers. For the CIFAR10 and CIFAR100 datasets, we designed three models: CNN, modified Resnet18 and Vgg11. The CNN model comprised two convolutional layers and three fully connected layers. The modified Resnet18 model replaced the original BatchNorm with GroupNorm to realise improved regularisation. To ensure a fair comparison, other baselines were evaluated using the CNN model on the EMNIST and modified Resnet18 datasets on the CIFAR10 and CIFAR100 datasets.

4.1.4. Baselines

We compared the proposed DFL-MKF algorithm to a series of baselines, including both centralised and decentralised FL methods. One of our approaches aimed to address the issue of data heterogeneity. Therefore, we first compared the proposed method to **FedAvg** (McMahan et al., 2017) and **Ditto** (Li, Hu, et al., 2021). In terms of data heterogeneity, FedAvg demonstrated a significant decrease in accuracy and Ditto mitigated this issue by introducing a regularisation term. We also considered the **DPSGD** (Lian et al., 2017) algorithm, a decentralised stochastic gradient method, as a baseline method for decentralised FL.

4.1.5. Implementation details

In this study, the evaluation criterion for all experiments was based on the average test accuracy of devices participating in federated training on the local test dataset. Unless otherwise specified, each edge device in the decentralised FL settings had a maximum of five neighbours, enabling each edge device to learn from the knowledge of approximately five neighbours simultaneously. The settings of the remaining experimental parameters are listed in Table 4. We used accuracy as a performance metric for experimental evaluation, as defined by Eq. (6).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \quad (6)$$

4.2. Experimental results

4.2.1. Performance metrics

In Fig. 5, we present the average test accuracy of edge devices on three different datasets with parameter $\beta = 0.3$ and various sampling fractions. Additionally, Fig. 4 illustrates the learning curves of various baselines under the same experimental settings.

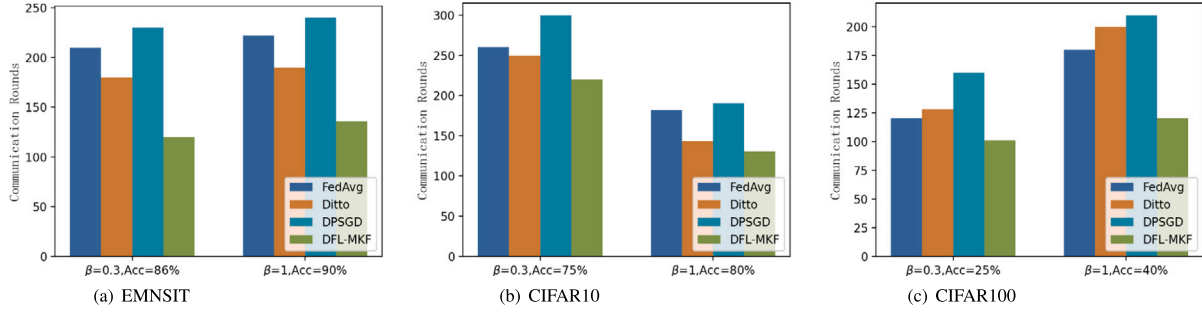


Fig. 5. Comparison of communication rounds among four algorithms when reaching the target test accuracy.

Analysing the experimental results, it is apparent that proposed algorithm DFL-MKF achieves higher accuracy, with the edge device outperforming those of other baseline approaches. Compared with FedAvg, the accuracy of our proposed algorithm on the EMNIST, CIFAR10 and CIFAR100 datasets is approximately improved by 0.92%, 2.72% and 2.06% respectively. It suggests that our algorithm can better adapt to data heterogeneity and mitigate the accuracy degradation issue caused by Non-IID data among local models compared to other baseline algorithms. There are two underlying reasons for this. On one hand, our approach addresses the heterogeneity of edge devices on hardware resources such as computing, storage, and bandwidth by designing specific models for each edge device. This not only facilitates better adaptation to the hardware heterogeneity but also enables the local models to better fit the local data. On the other hand, when multiple devices with heterogeneous data engage in communication, we employ knowledge transfer and fusion to assimilate the insights from multiple neighbours and enhance the performance of the local model. In contrast to other baseline methods employing simple parameter averaging, Our approach takes into account the influence of heterogeneous data from different devices by individually considering the model of each device. This strategy avoids dependence on a unified global model.

4.2.2. Comparison of communication efficiency

In order to verify the superiority of the proposed algorithm DFL-MKF in communication efficiency. We conduct experiments on three datasets, setting the target test accuracy under various values of β . For cases where $\beta=0.3$ and $\beta=1$, we set the target test accuracy on the EMNIST dataset is 86% and 90% respectively. Similarly, on the CIFAR10 dataset, the target test accuracy is set to 75% and 80% respectively, and on the CIFAR100 dataset, the target test accuracy is set to 25% and 40% respectively.

Fig. 5 evaluates the number of communications required for different algorithms to achieve the predefined target accuracy. Experimental results demonstrate that DFL-MKF requires the least number of communication rounds to achieve the predefined target test accuracy in the case of three datasets and two different degrees of heterogeneity. Compared with FedAvg, in the case of $\beta=0.3$, the number of communications of the proposed DFL-MKF on the EMNIST, CIFAR10 and CIFAR100 datasets is reduced by 42%, 15% and 16% respectively. In the case of $\beta=1$, the number of communications of the proposed DFL-MKF on three datasets is reduced by 40%, 27% and 30% respectively. It effectively proves that in heterogeneous data environments, the algorithm DFL-MKF can reduce communication rounds and improve federated training efficiency.

4.2.3. The impact of heterogeneity

The parameter β denotes the degree of Non-IID, the larger β , the smaller the Non-IID level of local data, on the contrary, the smaller β is, the higher the Non-IID level of local data. To investigate the

influence of different levels of Non-IID local data on the performance of edge devices, we perform experiments on three datasets using different values of $\beta \in \{0.1, 0.3, 1, 10, 100\}$.

From the experimental results in Fig. 6, we can observe that as β increases, the test accuracy continues to improve, the difference between different baselines becomes smaller, and DFL-MKF get higher performance than other baselines. This shows that when β is small, our algorithm can adapt to Non-IID data, and DFL-MKF can also achieve better performance than other baselines when β is large, suggesting the stability of our algorithm across diverse heterogeneous scenarios. It is because our algorithm via knowledge transfer and fusion can adapt to local data and learn from other neighbours, thus, mitigating the impact of heterogeneous data.

4.2.4. The impact of neighbours

Our algorithm DFL-MKF enhances the performance of the local model by integrating knowledge from multiple neighbours, and we also investigate the impact of the number of neighbours on the local model in our research. We perform experiments on three datasets using the proposed algorithm DFL-MKF, and set different numbers of neighbours S , including 1, 5, 10, and 20.

According to the experimental results in Fig. 7, we can see that with the number of neighbours increases, the performance of the local models improves gradually. There is a progressive increase in accuracy. In addition, we observe that the improvement in final test accuracy slows down and tends to stabilise when the number of neighbours exceeds 5. It demonstrates that by increasing the number of neighbours, our method can integrate more neighbour knowledge and introduce more samples and distribution scenarios, thereby achieving better performance on test samples and improving the generalisation ability of the local model. At the same time, we compensate for the differences among neighbours by fusing their knowledge and provide more comprehensive knowledge for edge devices.

However, increasing the number of neighbours also increases the communication costs among neighbours. Each neighbour needs to send its knowledge to the local model, which can add computational and communication burdens. And the edge device with few neighbours may result in a decrease in the accuracy of the local model. Therefore, we need to find a balance between model performance and communication costs. In our experiments, we found that the neighbour count of 5 achieves this balance.

4.2.5. The impact of α

In the proposed algorithm DFL-MKF, the parameter α is a hyper-parameter used to adjust the influence of the teacher model on knowledge transfer and fusion. It controls the weight of the teacher loss during the training process, and affect the guidance of teacher on the local model. Thus, we explore the effect of different α values on the local model by setting α within the range of $[0.1, 0.3, 0.5, 0.7]$.

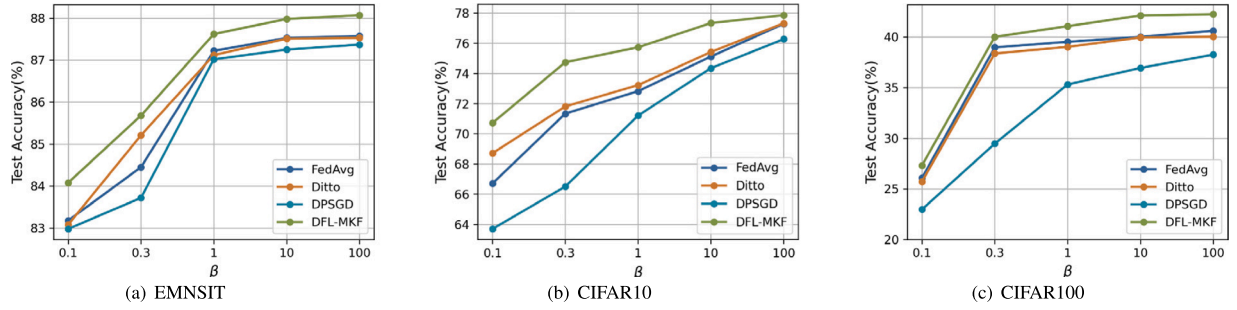


Fig. 6. Performance comparison of different heterogeneous levels on EMNIST, CIFAR10 and CIFAR100 datasets.

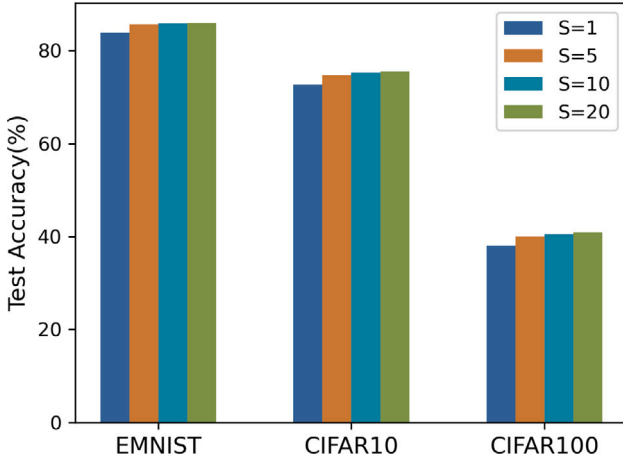


Fig. 7. Performance of edge devices under different number of neighbours S .

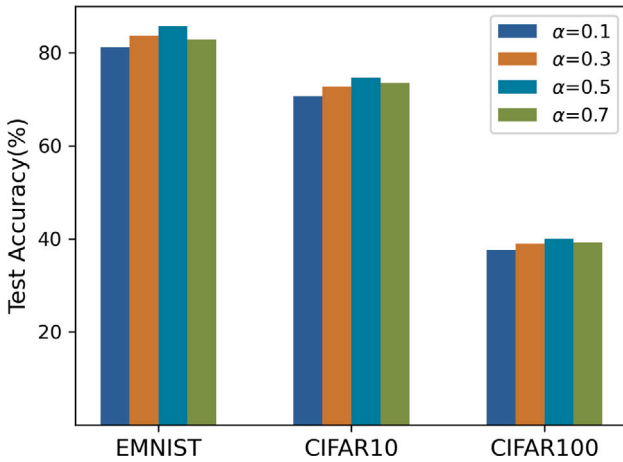


Fig. 8. Performance of edge devices under different hyper-parameter α .

From the experimental results shown in Fig. 8, we can observe that the test accuracy of the local model increases as the value of α increases before α reaches 0.5. However, the performance of the local model slightly decreases when α equals 0.7. It demonstrates that increasing the value of α within a certain range can enhance the performance of the local model, but excessively high values of α may lead to a decrease in the performance of the local model. It is attributed to the fact that increasing the value of α , the weight of the teacher loss in knowledge transfer is increased, which makes the local model more dependent on the knowledge of other neighbours, thereby improving performance of

Table 5

The table describes the test accuracy of edge devices under different local training epochs E .

Method	$E = 1$	$E = 5$	$E = 10$	$E = 20$
FedAvg	70.02	71.21	71.34	71.35
Ditto	71.02	71.03	71.81	71.24
DPSGD	65.83	66.41	66.52	66.37
DFL-MKF	73.86	74.58	74.73	74.52

local model. However, if the weight parameter α is too high, the local model may be too dependent on other neighbour models and lose the ability of the local model to learn. Conversely, if the parameter α is set too low, it may limit the flexibility of the local model, preventing it from capturing finer-grained knowledge.

4.2.6. The impact of local epochs E

Local epochs E represents the rounds that each edge device trains the local model in each federated communication. In order to explore the impact of different local epochs E on the performance of the local model, we set different E including 1, 5, 10, and 20 to perform experiments on four baselines. This experiment is conducted on the CIFAR10 dataset, the level of data heterogeneity β is set to 0.3, and other parameters adopt the default settings.

Table 5 shows the impact of different local training epochs E on the average test accuracy of four algorithms. It can be observed that the performance of the local model improves as the number of local epochs E increases. But after E exceeds 10, the accuracy of the local model starts to decline on four baselines. It indicates that the accuracy of the local model can be improved effectively by increasing the epochs of local training, because more local training allows the local model to better adapt to local data with different distributions. However, excessive the epochs of local training may lead to overfitting of the local data, thereby reducing the generalisation ability of the model on other devices. Additionally, increasing the epochs of local training also increases the computational resource consumption among edge devices. To strike a balance between model performance and device resources, in our experiments, we set the epochs of local training E to 10.

5. Conclusions

This study addressed the challenges associated with centralised server failure and data and system heterogeneity that hinder the application of FL to cognitive IoT while ensuring privacy protection. To address these challenges, we propose a decentralised FL framework called DFL-MKF that customises models for each edge device to accommodate the resources of heterogeneous devices. Employing knowledge fusion, the DFL-MKF model enabled edge devices to learn from the knowledge of multiple devices simultaneously. The proposed method effectively mitigated the impact of data heterogeneity on the performance of local models. Extensive experimental results demonstrate that the proposed DFL-MKF algorithm outperforms other baseline methods

in terms of test accuracy and communication efficiency while providing a privacy protection mechanism. The integration of KT and blockchain technology holds promise for establishing a decentralised FL framework for cognitive IoT applications. The proposed framework can further enhance user data privacy protection.

CRedit authorship contribution statement

Huanhuan Ge: Writing – original draft, Writing – review & editing, Conceptualization, Formal analysis, Investigation, Methodology, Software, Visualization. **Xingtao Yang:** Validation, Writing – review & editing. **Jinlong Wang:** Funding acquisition, Project administration, Writing – original draft, Writing – review & editing. **Zhihan Lyu:** Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by the Shandong Province Science and Technology Project, China (2023TSGC0509, 2022TSGC2234), Qingdao Science and Technology Plan Project, China (23-1-5-yqpy-2-qy, 24-1-2-qjyh-25-gx), and Open Topic Grants of Anhui Province Key Laboratory of Intelligent Building & Building Energy Saving, China, Anhui Jianzhu University, China (IBES2024KF08).

References

- Beltrán, E. T. M., Pérez, M. Q., Sánchez, P. M. S., Bernal, S. L., Bovet, G., Pérez, M. G., et al. (2022). Decentralized federated learning: Fundamentals, state-of-the-art, frameworks, trends, and challenges. *arXiv:2211.08413*.
- Beltrán, E. T. M., Pérez, M. Q., Sánchez, P. M. S., Bernal, S. L., Bovet, G., Pérez, M. G., et al. (2023). Decentralized federated learning: Fundamentals, state of the art, frameworks, trends, and challenges. *IEEE Communications Surveys & Tutorials*.
- Bhuyan, N., & Moharir, S. (2022). Multi-model federated learning. In *2022 14th international conference on cOMMunication systems & nETworks* (pp. 779–783). <http://dx.doi.org/10.1109/COMSNETS53615.2022.9668435>, *arXiv:2201.02582*.
- Chen, M., Herrera, F., & Hwang, K. (2018). Cognitive computing: architecture, technologies and intelligent applications. *IEEE Access*, 6, 19774–19783.
- Cohen, G., Afshar, S., Tapson, J., & Van Schaik, A. (2017). EMNIST: Extending MNIST to handwritten letters. In *2017 international joint conference on neural networks* (pp. 2921–2926). IEEE.
- Dai, R., Shen, L., He, F., Tian, X., & Tao, D. (2022). DisPFL: Towards communication-efficient personalized federated learning via decentralized sparse training. *arXiv:2206.00187*.
- Elmisery, A. M., Sertovic, M., & Gupta, B. B. (2017). Cognitive privacy middleware for deep learning mashup in environmental IoT. *IEEE Access*, 6, 8029–8041.
- Gabrielli, E., Pica, G., & Tolomei, G. (2023). A survey on decentralized federated learning. *arXiv preprint arXiv:2308.04604*.
- Gao, H., Thai, M. T., & Wu, J. (2023). When Decentralized Optimization Meets Federated Learning. *arXiv:2306.02570*.
- Gong, X., Sharma, A., Karanam, S., Wu, Z., Chen, T., Doermann, D., et al. (2021). Ensemble attention distillation for privacy-preserving federated learning. In *2021 IEEE/CVF international conference on computer vision* (pp. 15056–15066). IEEE, <http://dx.doi.org/10.1109/ICCV48922.2021.01480>.
- Grissa, M., Yavuz, A. A., & Hamdaoui, B. (2017). Location privacy preservation in database-driven wireless cognitive networks through encrypted probabilistic data structures. *IEEE Transactions on Cognitive Communications and Networking*, 3(2), 255–266.
- Han, Q., Lu, S., Wang, W., Qu, H., Li, J., & Gao, Y. (2024). Privacy preserving and secure robust federated learning: A survey. *Concurrency Computations: Practice and Experience*, Article e8084.
- He, C., Annaram, M., & Avestimehr, S. (2020). Group knowledge transfer: Federated learning of large CNNs at the edge. In *Advances in neural information processing systems*: vol. 33, (pp. 14068–14080). Curran Associates, Inc..
- Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network. *arXiv:1503.02531*.
- Islam, T. U., Ghasemi, R., & Mohammed, N. (2022). Privacy-preserving federated learning model for healthcare data. In *2022 IEEE 12th annual computing and communication workshop and conference* (pp. 0281–0287). IEEE.
- Kairouz, P., McMahan, H., Avent, B., Bellet, A., Bennis, M., Bhagoji, A., et al. (2019). Advances and open problems in federated learning. Now Publishers, Inc.,
- Karimireddy, S. P., Kale, S., Mohri, M., Reddi, S., Stich, S., & Suresh, A. T. (2020). SCAFFOLD: Stochastic controlled averaging for federated learning. In *Proceedings of the 37th international conference on machine learning* (pp. 5132–5143). PMLR.
- Khan, L. U., Alsenwi, M., Yaqoob, I., Imran, M., Han, Z., & Hong, C. S. (2020). Resource optimized federated learning-enabled cognitive internet of things for smart industries. *IEEE Access*, 8, 168854–168864.
- Krizhevsky, A., Hinton, G., et al. (2009). *Learning multiple layers of features from tiny images*. Toronto, ON, Canada.
- Li, T., Hu, S., Beirami, A., & Smith, V. (2021). Ditto: Fair and robust federated learning through personalization. In *Proceedings of the 38th international conference on machine learning* (pp. 6357–6368). PMLR.
- Li, F., Lam, K.-Y., Li, X., Sheng, Z., Hua, J., & Wang, L. (2019). Advances and emerging challenges in cognitive internet-of-things. *IEEE Transactions on Industrial Informatics*, 16(8), 5489–5496.
- Li, C., Li, G., & Varshney, P. K. (2022). Decentralized federated learning via mutual knowledge transfer. *IEEE Internet of Things Journal*, 9(2), 1136–1147. <http://dx.doi.org/10.1109/JIOT.2021.3078543>.
- Li, C., Li, C., Zhao, Y., Zhang, B., & Li, C. (2021). An efficient multi-model training algorithm for federated learning. In *2021 IEEE global communications conference* (pp. 1–6). <http://dx.doi.org/10.1109/GLOBECOM46510.2021.9685230>.
- Li, T., Sahu, A. K., Talwalkar, A., & Smith, V. (2020). Federated learning: Challenges, methods, and future directions. *IEEE Signal Processing Magazine*, 37(3), 50–60. <http://dx.doi.org/10.1109/MSP.2020.2975749>, *arXiv:1908.07873*.
- Li, S., Zhou, T., Tian, X., & Tao, D. (2022). Learning to collaborate in decentralized learning of personalized models. In *2022 IEEE/CVF conference on computer vision and pattern recognition* (pp. 9756–9765). IEEE, <http://dx.doi.org/10.1109/CVPR52688.2022.00954>.
- Lian, X., Zhang, C., Zhang, H., Hsieh, C.-J., Zhang, W., & Liu, J. (2017). Can decentralized algorithms outperform centralized algorithms? A case study for decentralized parallel stochastic gradient descent. In *Advances in neural information processing systems*: vol. 30, Curran Associates, Inc..
- Liu, Y., Kang, Y., Zou, T., Pu, Y., He, Y., Ye, X., et al. (2024). Vertical federated learning: Concepts, advances, and challenges. *IEEE Transactions on Knowledge and Data Engineering*.
- Liu, X., Li, Y., Shao, Y., & Wang, Q. (2022). Sparse federated learning with hierarchical personalized models. *arXiv:2203.13517*.
- Lu, Z., Pan, H., Dai, Y., Si, X., & Zhang, Y. (2024). Federated learning with non-IID data: A survey. *IEEE Internet of Things Journal*.
- Ma, X., Zhang, J., Guo, S., & Xu, W. (2022). Layer-wise model aggregation for personalized federated learning. In *2022 IEEE/CVF conference on computer vision and pattern recognition* (pp. 10082–10091). IEEE, <http://dx.doi.org/10.1109/CVPR52688.2022.00985>.
- McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data. *arXiv:1602.05629*.
- Muhammad, A., Lin, K., Gao, J., & Chen, B. (2022). Robust multi-model personalized federated learning via model distillation. In Y. Lai, T. Wang, M. Jiang, G. Xu, W. Liang, & A. Castiglione (Eds.), *Algorithms and architectures for parallel processing*: vol. 13157, (pp. 432–446). Springer International Publishing, http://dx.doi.org/10.1007/978-3-030-95391-1_27.
- Rasha, A.-H., Li, T., Huang, W., Gu, J., & Li, C. (2023). Federated learning in smart cities: Privacy and security survey. *Information Sciences*.
- Schilero, F., Moustafa, N., & Beheshti, A. (2020). Cognitive privacy: AI-enabled privacy using EEG signals in the internet of things. In *2020 IEEE 6th international conference on dependability in sensor, cloud and big data systems and application* (pp. 73–79). IEEE.
- Shi, H., Zhang, J., Fan, S., Ma, R., & Guan, H. (2023). pFedEff: An efficient and personalized federated cognitive learning framework in multi-agent systems. *IEEE Transactions on Cognitive and Developmental Systems*.
- Tan, A. Z., Yu, H., Cui, L., & Yang, Q. (2022). Towards personalized federated learning. *IEEE Transactions on Neural Networks and Learning Systems*, 1–17. <http://dx.doi.org/10.1109/TNNLS.2022.3160699>, *arXiv:2103.00710*.
- Vellingiri, J., Kalaivanan, K., Gopinath, M., Gobinath, C., Subramaniam, P. R., & Rangarajan, S. (2023). Strategies for classifying water quality in the cauvery river using a federated learning technique. *International Journal of Cognitive Computing in Engineering*, 4, 187–193.
- Wang, G. (2017). Data-driven granular cognitive computing. In *Rough sets: international joint conference, IJCRS 2017, Olsztyn, Poland, July 3–7, 2017, proceedings, part I* (pp. 13–24). Springer.
- Witt, L., Heyer, M., Toyoda, K., Samek, W., & Li, D. (2023). Decentral and incentivized federated learning frameworks: A systematic literature review. *IEEE Internet of Things Journal*, 10(4), 3642–3663. <http://dx.doi.org/10.1109/JIOT.2022.3231363>.
- Wu, Q., Ding, G., Xu, Y., Feng, S., Du, Z., Wang, J., et al. (2014). Cognitive internet of things: a new paradigm beyond connection. *IEEE Internet of Things journal*, 1(2), 129–143.
- Yang, J., Zhou, Y., Wen, W., Zhou, J., & Zhang, Q. (2023). Asynchronous hierarchical federated learning based on bandwidth allocation and client scheduling. *Applied Sciences*, 13(20), 11134.

- Yi, L., Gang, W., & Xiaoguang, L. (2022). QSFL: A two-level uplink communication optimization framework for federated learning. In *Proceedings of the 39th international conference on machine learning* (pp. 25501–25513). PMLR.
- Zhang, J., Guo, S., Ma, X., Wang, H., Xu, W., & Wu, F. (2021). Parameterized knowledge transfer for personalized federated learning. In *Advances in neural information processing systems: vol. 34*, (pp. 10092–10104). Curran Associates, Inc..
- Zhang, W., Wang, X., Zhou, P., Wu, W., & Zhang, X. (2021). Client selection for federated learning with non-IID data in mobile edge computing. *IEEE Access*, 9, 24462–24474. <http://dx.doi.org/10.1109/ACCESS.2021.3056919>.
- Zhu, Z., Hong, J., & Zhou, J. (2021). Data-free knowledge distillation for heterogeneous federated learning. *arXiv:2105.10056*.
- Zong, R., Qin, Y., Wu, F., Tang, Z., & Li, K. (2024). Fedcs: Efficient communication scheduling in decentralized federated learning. *Information Fusion*, 102, Article 102028.